Slip No. 1

Q1.Create an account on Google Cloud Platform (GCP) using either Windows or Linux. Identify and list the services available under the Always Free tier.

Ans –

Part A: Steps to Create a GCP Account

Step 1: Open Google Cloud Start your Windows or Linux computer. Open a web browser. Search for Google Cloud Console. Open the official Google Cloud website. Click Get started for free or Start free.

Step 2: Sign in Sign in using your Google/Gmail account. If you do not have a Google account, create one first. Accept the required terms and conditions.

Step 3: Enter Account Details Select your country. Choose the appropriate account type. Enter the required personal or business information. Continue to the next step.

Step 4: Set Up Billing Google Cloud may ask you to create a billing account. Enter the required payment information if requested. Follow the on-screen verification instructions. Google Cloud's free trial and billing terms can change, so check the current terms displayed during sign up.

Step 5: Open Google Cloud Console After completing registration, open the Google Cloud Console. The Cloud Console provides access to different GCP services. Create or select a Google Cloud project.

Step 6: Verify the Account Check that the Cloud Console opens successfully. Open the Billing section to view billing information. Open the IAM & Adminsection to view project permissions. Your GCP account is now ready to use.
Part B: Always Free Tier Services
Google Cloud provides selected products with monthly usage limits under its Always Free program. The exact limits and eligible regions can change, so students should verify them on Google's current pricing page.
Common Always Free offerings include:

GCP Service	Example Always Free Allowance
Compute Engine	Limited usage of certain eligible VM configurations in selected US regions
Cloud Storage	Limited monthly storage, operations and network usage in eligible locations
BigQuery	Limited monthly query processing and storage
Cloud Run	Limited monthly requests, CPU and memory usage
Cloud Functions	Limited monthly invocations and compute resources
Firestore	Limited storage, reads, writes and deletes
Pub/Sub	Limited monthly message/data usage
Google Kubernetes Engine (GKE)	Limited resources/usage under applicable free-tier conditions

Q2. Install a web server (such as Apache or NGINX) on your virtual machine and verify its functionality by accessing it through a web browser.

Ans –

Steps to Install Apache Web Server on Linux Virtual Machine

1.Start the Linux virtual machine and connect to it using SSH or the virtual machine terminal.
2.Update the package list using the command: sudo apt update
3.Install Apache web server using the command: sudo apt install apache2 -y
4.Start the Apache service using the command: sudo systemctl start apache2
5.Enable Apache to start automatically when the VM is restarted using: sudo systemctl enable apache2
6.Check the status of Apache using: sudo systemctl status apache2
7. If the virtual machine is hosted on a cloud platform, allow HTTP traffic on port 80 in the VM's firewall/security rules.
8. Find the public IP address of the virtual machine.
9. Open a web browser on your computer.
10. Enter the public IP address in the browser, for example: http://<public-ip>
11. The default Apache web page should appear in the browser.

Write This Code
sudo nano /var/www/html/index.html—use for find index.html file
Ctrl+k— to delete all data from index file clear
Write This code –

```
  GNU nano 8.7.1
<!DOCTYPE html>
<html>
<head>
        <title>My Web Server</title>
</head>
<body>
        <h1>Welcome to my web server!</h1>
        <p>My Web Server Is Running!</p>
</body>
</html>
```

Click ctrl+x to save file and click y and click enter.
Go to firefox and type localhost.

Slip No. 2

Q1. Create a SQL database on Google Cloud Platform (GCP) using either Windows or Linux. Write step-by-step instructions for the process.

Ans –

SQL database using GCP using windows
Step 1: Sign in to GCP Console

1. Open your browser and go to: **https://console.cloud.google.com**
2. Sign in using your Google account.
3. At the top, click on the project dropdown and select or create a new project.

Step 2: Enable the Cloud SQL API

1. In the top search bar, type Cloud SQL.
2. Click on SQL from the results.
3. If prompted, click Enable API.

Step 3: Create a Cloud SQL Instance

1. Click “Create Instance”.
2. Choose your database engine: o MySQL o PostgreSQL o SQLServer
3. Fill in instance details:
   - o Instance ID: e.g., my-sql-instance
   - o Root password: Set a strong password
   - o Region: Choose a nearby location
   - o Machine Type: Select db-f1-micro (eligible for free tier)
   - 4.Click Create and wait 3–5 minutes for setup.

Step 4: Create a New Database

1. After the instance is ready, click on its name.
2. Go to the Databases tab.
3. Click “Create Database”.
4. Enter the database name (e.g., student_db).
5. Leave other settings as default and click Create.

Step 5: Connect to Your SQL Database
You can connect using:Cloud Shell (built-in terminal):

1. Click the terminal icon at the top-right of the GCP Console.
2. Run the command: gcloud sql connect my-sql-instance--user=root

Q2. Demonstrate how to create a virtual machine using the IaaS service of your chosen cloud provider (GCP, AWS,orAzure). Include the step-by-step process. Give examples from GCP, AWS, and Azure
Ans –

Infrastructure as a Service (IaaS) is a cloud computing service model that provides virtualised computing resources such as:

- Virtual machines
- CPU
- Memory
- Storage
- Networking

Examples of IaaS services are:

- Cloud Provider = IaaS Services
- GCP Compute = Engine
- AWS = Amazon EC2
- Microsoft Azure = Azure Virtual Machines

Creating a VM in AWS

1) Open the AWS Management Console.
2) Sign in to your AWS account.
3) Search for and open EC2.
4) Click Launch Instance.
5) Enter a name for the instance.
6) Select an AMI (Amazon Machine Image).
7) Choose an operating system such as:
    - Amazon Linux
    - Ubuntu
    - Windows Server
8) Select an appropriate Instance Type.
9) Select or create a Key Pair.
10) Configure the Network settings.
11) Configure the Security Group.
12) Allow required ports such as:
    - SSH – Port 22 for Linux
    - RDP – Port 3389 for Windows
13) Configuration Storage
14) Review the configuration
15) Click Launch Instance
16) Wait until the instance reaches the Running state.
17) Select the instance & use Connect to connect to the VM.

Slip No. 3

Q1. Create and configure a custom firewall rule to enable HTTP and SSH traffic to your virtual machine.
Ans –

Steps to Create a Custom Firewall Rule in GCP

1. Login to the Google Cloud Console.
2. Select the required GCP project.
3. Go to VPC Network → Firewall.
4. Click 'Create Firewall Rule'.
5. Enter a rule name, for example:
allow-http-ssh.
6. Select the required network, such as default.
7. Under Direction of traffic, select
'Ingress'.
8. Under Action on match, select
    a) 'Allow'.
9. Select the target VM or use a target network tag such as
    a) web-server.
10. Under Source IPv4 ranges, enter
    a) 0.0.0.0/0.
11. Under Protocols and ports, select specified protocols and ports.
12. Select TCP and enter 22,80.
13. Port 22 is used for SSH and Port 80 is used for HTTP.
14. Click 'Create'.

Configure the VM Network Tag

1. Go to Compute Engine → VM instances.
2. Select the VM and click 'Edit'.
3. Under Network tags, add web-server.
4. Click 'Save'.

Verify the Firewall Rule

1. Connect to the VM using SSH.
2. Start Apache or another web server.
3. Open http://<external-IP> in a browser.
4. Verify that the web server page is displayed

Q2. What are the steps to assign a static IP address to a virtual machine on Google Cloud Platform (GCP)?
Ans –

Steps to Reserve and Assign a Static IP Address

1. Login to the Google Cloud Console.
2. Select the required GCP project.
3. Go to: VPC Network → IP addresses.
4. Click 'Reserve external static IP address'.
5. Enter a name, for example: my-static-ip.
6. Select the required region and IPv4.
7. Click 'Reserve'.
8. Go to Compute Engine → VM instances.
9. Select the VM and click 'Edit'.
10. Go to the Network interfaces section.
11. Under External IPv4 address, select the reserved static IP.
12. Click 'Done' and then 'Save'.
13. Open the VM details and verify the assigned static IP.

Verify the Static IP

1. Check the external IP shown for the VM.
2. If a web server is installed, open http://<static-IP> in a browser.
3. Verify that the web page is accessible using the static IP

Slip No. 4

Q1. Launch an EC2 instance on AWS using the Free Tier. Specify the configuration you selected, including the operating system, region, and instance type.
Ans –

Launch as EC2 instance on AWS using the Free Tier

1)Open the Amazon Web Services Management Console and sign in to your AWS account.
2)Search for EC2 and open the EC2 Dashboard.
3)Click Launch Instance.
4)Enter the instance name, for example: My-Free-Tier-EC2

Select the following configuration:

Configuration	Selected Option
Operating System	Ubuntu Server
Region	Asia Pacific (Mumbai) – ap-south-1
Instance Type	t2.micro
Architecture	64-bit (x86)

5)Under Key pair, create a new key pair, for example: my-ec2-key
6)Download and save the .pem file safely.
7)Under Network settings, allow SSH (Port 22) so that the instance can be accessed remotely.
8)Review the configuration and click Launch Instance. After a few moments, the instance status becomes Running.
9)Select the instance and note its Public IPv4 address.
10)The instance can now be connected using SSH:

ssh -i "my-ec2-key.pem" ubuntu@<Public-IP>

Selected Configuration
OS: Ubuntu Server
Region: Asia Pacific (Mumbai)
Instance Type: t2.micro
Purpose: Free Tier-eligible EC2 virtual machine

Q2. Explain the Three Primary Cloud Service Models —Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) — with Relevant Examples and Real World Use Cases.
Ans –

IaaS – Infrastructure as a Service

IaaS provides **virtual computing resources** such as virtual machines, storage, and networking through the internet.
Example: AWS EC2
Real-world use case:
A company can create a virtual server on AWS EC2 to host its website without purchasing a physical server.
Main advantages:
No need to purchase physical hardware.
Resources can be increased or decreased as required.
Users have control over the operating system and applications.

PaaS – Platform as a Service

PaaS provides a **ready-to-use platform** for developing, testing, and deploying applications. The cloud provider manages the infrastructure and operating system.
Example: Google App Engine
Real-world use case:
A developer can deploy a Python web application using Google App Engine without manually managing the server or operating system.
Main advantages:
Faster application development.
No need to manage servers.
Easy application deployment and scaling.

SaaS – Software as a Service

SaaS provides **ready-to-use software applications** over the internet. Users do not need to install or maintain the software on their computers.
Example: Google Docs
Real-world use case:
Students and teachers can use Google Docs to create, edit, and share documents through a web browser.
Main advantages:
No installation required.
Accessible from anywhere with an internet connection.
Software updates and maintenance are handled by the provider.

Slip No. 5

Q1. Create an account and explore a Software as a Service (SaaS) product like Google Docs, Microsoft 365, or Salesforce. Analyse and evaluate the key features and overall functionality of the platform.
Ans –

Steps to Create an Account and Explore Google Docs :-
1)Open a web browser and visit Google Docs.
2)Sign in using a Google account. If you do not have an account, create a Google account.
3)Open **Google Docs** and select **Blank Document**.
4)Create a sample document and explore the available features.
5)Try editing text, inserting images, creating tables, and formatting the document.
6)Use the **Share** option to share the document with another user.
7)Give the user appropriate permission such as **Viewer, Commenter, or Editor**.
8)Add a comment to test the collaboration feature.
9)Check the **Version History** to view previous changes.
10)Close and reopen the document to verify that the changes are automatically saved.

Key Features of Google Docs
1)Online Document Creation
Users can create and edit documents directly through a web browser without installing traditional word-processing software.

2)Cloud Storage
Documents are stored online and can be accessed from different devices using the user's account.

3)Real-Time Collaboration
Multiple users can work on the same document at the same time.

4)Sharing and Permissions
Documents can be shared with other users. Different access permissions such as **Viewer, Commenter, and Editor** can be provided.

5)Automatic Saving
Changes are automatically saved to the cloud, reducing the risk of losing work.

Q2. Describe Two Practical Use Cases Where Cloud Databases Offer Significant Advantages.
Ans –

Two Practical Use Cases of Cloud Databases :-

1)E-Commerce Application –

Cloud databases are useful for storing customer details, product information, orders, and payment records.

- Data can be accessed from anywhere through the internet.
- The database can handle a large number of users during peak sales.
- Storage and computing resources can be increased when required.
- Backup and recovery facilities help protect important business data.

Example: An online shopping website can use a cloud database to store customer orders and product information.

2)Banking and Financial Applications –

Cloud databases can store customer accounts, transactions, loan details, and payment information.

- Data can be accessed by authorized users from different branches.
- Cloud databases provide scalability for handling a large number of transactions.
- Backup and recovery help prevent data loss.
- Security and access-control features help protect sensitive information.

Example: A bank can use a cloud database to manage customer accounts and online banking transactions.

Slip No. 6

Q1. Launch a VM(EC2 on AWS/ComputeEngine on GCP/AzureVM).Identify it as an example of IaaS.
Ans –

Launch a VM on AWS and Identify It as IaaS :-

For this practical, **AWS EC2** is selected as the cloud service.

a) Open the Amazon Web Services Management Console and sign in.
b) Search for **EC2** and open the **EC2 Dashboard**.
c) Select the required **AWS Region**, for example: **Asia Pacific (Mumbai) –** ap-south-1
d) Click **Launch Instance**.
e) Enter an instance name: My-IaaS-Instance
f) Select the following configuration:
    I. **Operating System:** Ubuntu Server
    II. **Instance Type:** t2.micro (where Free Tier eligible)
    III. **Architecture:** 64-bit
    IV. **Storage:** Default storage

g) Create a **Key Pair** and download the .pem file.
h) In **Network Settings**, allow **SSH (Port 22)** for remote connection.
i) Click **Launch Instance**.
Wait until the instance state becomes **Running**.
j) Select the instance and note its **Public IPv4 address**. The VM can be connected using SSH:

ssh -i "My-IaaS-Instance.pem" ubuntu@<Public-IP>

Why EC2 is an Example of IaaS

AWS EC2 is an example of Infrastructure as a Service (IaaS) because AWS provides the basic computing infrastructure as a virtual machine.
The user can manage:

- Operating system
- Applications
- Software
- Storage configuration
- Network settings

Q2. Describe Two Practical Use Cases Where Cloud Databases Offer Significant Advantages.
Ans –

Two Practical Use Cases of Cloud Databases :-

1)E-Commerce Application –

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Example: A bank can use a cloud database to manage customer accounts and online banking transactions.

Slip No. 7

Q1. Describe Two Practical Use Cases Where Cloud Databases Offer Significant Advantages.
Ans –

Two Practical Use Cases of Cloud Databases :-

1)E-Commerce Application –

Cloud databases are useful for storing customer details, product information, orders, and payment records.

- Data can be accessed from anywhere through the internet.
- The database can handle a large number of users during peak sales.
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Example: An online shopping website can use a cloud database to store customer orders and product information.

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- Data can be accessed by authorized users from different branches.
- Cloud databases provide scalability for handling a large number of transactions.
- Backup and recovery help prevent data loss.
- Security and access-control features help protect sensitive information.

Example: A bank can use a cloud database to manage customer accounts and online banking transactions.

Q2. Install the Apache Web Server on a newly launched Linux virtual machine (VM). Provide the step-by-step instructions for the installation process.
Ans –

Install Apache Web Server on a Linux VM : -

For this practical, assume the Linux VM is an **Ubuntu Server on AWS EC2**.

Launch an **Ubuntu EC2 instance** on AWS. In the **Security Group**, allow:

- **SSH – Port 22**
- **HTTP – Port 80**

Connect to the VM using SSH:

ssh -i "my-key.pem" ubuntu@<Public-IP>

A. Update the package list: sudo apt update

B. Install the Apache Web Server: sudo apt install apache2 -y

C. Check the Apache service status: sudo systemctl status apache2

If Apache is running, you will see **active (running)**.
If Apache is not running, start it using: sudo systemctl start apache2

Enable Apache to start automatically when the VM boots:
sudo systemctl enable apache2

Open a web browser and enter: http://<Public-IP>

The **Apache2 Ubuntu Default Page** should appear.

Slip No. 8

Q1. Explain the Three Primary Cloud Service Models —Infrastructure as a Service(IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) — with Relevant Examples and Real-World Use Cases.
Ans –

IaaS – Infrastructure as a Service

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PaaS provides a **ready-to-use platform** for developing, testing, and deploying applications. The cloud provider manages the infrastructure and operating system.
Example: Google App Engine
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A developer can deploy a Python web application using Google App Engine without manually managing the server or operating system.
Main advantages:
Faster application development.
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SaaS – Software as a Service

SaaS provides **ready-to-use software applications** over the internet. Users do not need to install or maintain the software on their computers.
Example: Google Docs
Real-world use case:
Students and teachers can use Google Docs to create, edit, and share documents through a web browser.
Main advantages:
No installation required.
Accessible from anywhere with an internet connection.
Software updates and maintenance are handled by the provider.

Q2. What are the pricing options available for compute services across different cloud providers, and how do their pricing tools differ?
Ans –

Pricing Option	**AWS**	**Azure**	**GCP**
Pay-as-you-go	On-Demand Instances	Pay-as-you-go VMs	On-demand Compute Engine
Reserved or Committed Usage	Reserved Instances / Savings Plans	Reserved VM Instances / Savings Plans	Committed Use Discounts
Spot / Preemptible Instances	Spot Instances	Spot VMs	Spot VMs
Free Tier	Free Tier where eligible	Free services/credits where eligible	Free Tier/credits where eligible

Difference Between Pricing Tools

AWS Pricing Calculator

- Used to estimate the cost of AWS services.
- Allows users to select services such as EC2, storage, and databases.
- Provides an estimated monthly cost.

Azure Pricing Calculator

- Used to estimate Azure service costs.
- Allows selection of VM size, operating system, region, and usage.
- Provides estimated costs based on the selected configuration.

Google Cloud Pricing Calculator

- Used to estimate Google Cloud service costs.
- Allows users to select Compute Engine machine type, region, usage, and other resources.
- Provides estimated costs for the selected configuration.

Slip No. 9

Q1. How do you deploy basic cloud storage (object, block, and file) and launch managed databases across AWS?
Ans –

1)Object Storage – Amazon S3
Amazon S3 is used to store files and objects such as images, videos, documents, and backups.
Steps:

1. Sign in to the AWS Management Console.
2. Search for **S3** and open the S3 service.
3. Click **Create bucket**.
4. Enter a unique bucket name.
5. Select the required AWS Region.
6. Keep **Block Public Access** enabled for secure private storage.
7. Click **Create bucket**.
8. Open the bucket and click **Upload**.
9. Select a file and click **Upload**.

The file is now stored as an object in the S3 bucket.

2)Block Storage – Amazon EBS
Amazon EBS provides **block storage** that can be attached to an EC2 virtual machine.
Steps:

1. Open the **EC2 Dashboard**.
2. Select **Volumes** under Elastic Block Store.
3. Click **Create volume**.
4. Select the required volume type, such as **gp3**.
5. Specify the required storage size.
6. Select the same Availability Zone as the EC2 instance.
7. Click **Create volume**.
8. Select the volume and choose **Actions → Attach volume**.
9. Select the required EC2 instance.
10. The EBS volume is now attached to the VM.

Use: EBS is useful for operating systems, applications, and databases that require persistent disk storage.

3)File Storage – Amazon EFS
Amazon EFS provides a **shared file system** that can be accessed by multiple EC2 instances.
Steps:

1. Search for **EFS** in the AWS Console.
2. Click **Create file system**.
3. Select the required VPC.
4. Configure the Availability Zones and security settings.
5. Click **Create**.
6. Select the created file system and open **Attach**.
7. Follow the provided mount instructions on the Linux EC2 instance.

Use: EFS is useful when multiple servers need to access the same files.

Managed Database – Amazon RDS
Amazon RDS is a managed relational database service. AWS manages tasks such as database infrastructure, backups, and maintenance.
Steps:

1. Search for RDS in the AWS Console.
2. Click Create database.
3. Select Standard Create.
4. Choose a database engine such as MySQL.
5. Select an eligible instance class according to your account and requirements.
6. Enter the database name, username, and password.
7. Configure the VPC and security group.
8. Configure storage according to the requirement.
9. Click Create database.
10. After the database becomes available, copy its endpoint and use it to connect to the database.

Q2. What are the pricing options available for compute services across different cloud providers, and how do their pricing tools differ?
Ans –

Pricing Option	**AWS**	**Azure**	**GCP**
Pay-as-you-go	On-Demand Instances	Pay-as-you-go VMs	On-demand Compute Engine
Reserved or Committed Usage	Reserved Instances / Savings Plans	Reserved VM Instances / Savings Plans	Committed Use Discounts
Spot / Preemptible Instances	Spot Instances	Spot VMs	Spot VMs
Free Tier	Free Tier where eligible	Free services/credits where eligible	Free Tier/credits where eligible

Difference Between Pricing Tools

AWS Pricing Calculator

- Used to estimate the cost of AWS services.
- Allows users to select services such as EC2, storage, and databases.
- Provides an estimated monthly cost.

Azure Pricing Calculator

- Used to estimate Azure service costs.
- Allows selection of VM size, operating system, region, and usage.
- Provides estimated costs based on the selected configuration.

Google Cloud Pricing Calculator

- Used to estimate Google Cloud service costs.
- Allows users to select Compute Engine machine type, region, usage, and other resources.
- Provides estimated costs for the selected configuration.

Slip No. 10

Q1. Explain the concepts of versioning, encryption, and data recovery in the context of cloud storage. Describe how these features work and their importance in ensuring data integrity, security, and availability.
Ans –

1)Versioning:-
Versioning means keeping multiple versions of the same file or object.

How it works:
1. A file is uploaded to cloud storage.
2. When the file is modified or replaced, the previous version is retained.
3. Each version can have a different version ID.
4. If a file is accidentally deleted or modified, an older version can be restored.

Example:
If report.pdf is accidentally deleted, the previous version can be recovered from the stored versions.
Importance:
- Prevents accidental data loss.
- Helps recover accidentally deleted or modified files.
- Maintains **data integrity**.

2)Encryption:-
Encryption converts data into an unreadable format so that unauthorized users cannot understand it.

There are two common types:
1. Encryption at Rest: Protects data while it is stored in the cloud.
2. Encryption in Transit: Protects data while it is being transferred between the user and cloud storage.

How it works:
- Data is converted into encrypted form using an encryption algorithm/key.
- The encrypted data is stored or transferred.
- An authorized user or service uses the appropriate key to decrypt the data.

Importance:
- Protects confidential information.
- Prevents unauthorized access.
- Improves data security and privacy.

3)Data Recovery:-
Data recovery means restoring data after accidental deletion, corruption, hardware failure, or other problems.

How it works:

1. Data is regularly backed up or replicated.
2. Copies may be stored in different locations.
3. If the original data is lost or damaged, a backup copy can be restored.
4. The application can continue working with the recovered data.

Importance:

- Protects against data loss.
- Helps restore services after failures.
- Ensures data availability.
- Supports business continuity.

Q2. Create a new Linux-based virtual machine on your preferred cloud platform (e.g., GCP, AWS, Azure), then install and configure the Apache Web Server on the newly launched VM.
Ans –

Create a Linux VM on AWS and Install Apache Web Server

For this practical, **AWS EC2** is selected as the cloud platform.

1)Launch the Linux VM:-

1. Sign in to the **AWS Management Console**.
2. Search for **EC2** and open the EC2 Dashboard.
3. Select the region **Asia Pacific (Mumbai) –** ap-south-1.
4. Click **Launch Instance**.
5. Enter the instance name: Apache-Web-Server
6. Select:
   - **Operating System:** Ubuntu Server
   - **Instance Type:** t2.micro where eligible
   - **Architecture:** 64-bit
7. Create a new **Key Pair** and download the .pem file.
8. Under **Network Settings**, allow:
   - SSH – Port 22
   - **HTTP – Port 80**
9. Click **Launch Instance**.
10. Wait until the instance state becomes **Running**.
11. Copy the instance's **Public IPv4 address**.

2)Connect to the Linux VM:-

- Open a terminal or PowerShell and run:

ssh -i "my-key.pem" ubuntu@<Public-IP>

Replace <Public-IP> with the public IP address of your EC2 instance.

3)Install Apache:-

- Update the package list: sudo apt update
- Install Apache: sudo apt install apache2 -y

4)Start and Configure Apache:-

- Check the Apache service: sudo systemctl status apache2

- If it is not running, start it: sudo systemctl start apache2

Enable Apache to start automatically after reboot:

- sudo systemctl enable apache2

5)Test the Web Server
Open a web browser and enter: http://<Public-IP>